6 **(a)** On Fig. 6.1, sketch the current–voltage graph of a filament lamp and explain its shape.



Fig. 6.1

explanation .....

.....

[3]

(b) Fig. 6.2 shows an electric circuit.

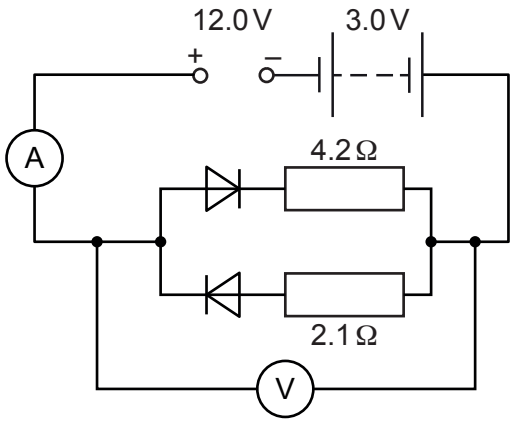


Fig. 6.2

(i) Calculate the reading on the voltmeter.

voltmeter reading = ... [2]

(ii) Calculate the current in the 4.2 Ω resistor.

current = ... [2]

(iii) Determine the current in the 2.1 Ω resistor.

current = ... [1]

(iv) Determine the reading on the ammeter.

ammeter reading = ... [1]

(v) Calculate the electrical power transferred in the 4.2 Ω resistor.

power = ... [2]

[Total: 11]